(Please write your Enrolment no. immediately) Enrolment No._____

SUPPLEMENTARY MID TERM EXAMINATION

B.Tech Programmes (Under the Aegis of USICT)

IV Semester, March 2025

Paper Code: CIC-210 **Subject: Database Management System**

Time: 1.5 Hrs **Max Marks: 30**

Note: Attempt Question 1, which is compulsory, and any two more questions from the remaining.

Q. No.		Question	Max Marks	CO(s)
1.	a)	Explain the difference between Physical data independence and Logical data independence.	2.5	CO1
	b)	Who is a Database Administrator? Explain the responsibilities of a database administrator (DBA).	2.5	CO1
	c)	Explain the concept of aggregation with an example.	2.5	CO1
	d)	What are the different types of database languages? Explain with examples.	2.5	CO2
2.	a)	Define the terms entity, attribute, and relationship with suitable examples.	5	CO1
	b)	Draw an ER diagram for a Library Management System, considering entities like Books, Members, Librarians, and Transactions. Clearly indicate attributes and relationships.	5	CO1
3.		Consider the following relational schema: Customers (Customer_ID, Name, Address, Phone) Accounts (Account_No, Customer_ID, Account_Type, Balance) Transactions (Transaction_ID, Account_No, Transaction_Type, Amount, Date) Write SQL Command and Relational algebra expression for each of following queries. i) Find the names and account numbers of customers who have a savings account. ii) Retrieve the total balance of all accounts in the bank. iii) Find customers who have a balance of more than ₹50,000. iv) Find the total balance for each account type. v) Find account types where the total balance exceeds ₹1,00,000.	2*5=10	CO2
4.	a)	Differentiate between Primary Key and Foreign Key with an example.	5	CO2
	b)	Explain Left Outer Join, Right Outer Join and Full Outer Join with an example.	5	CO2